



MCS3301 – Control Engineering I

Dr. Ado Haruna
Dept. of Mechatronics Engineering
Bayero University, Kano
aharuna.mct@edu.buk.ng

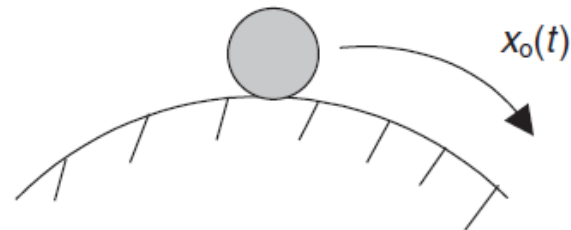


System Stability

- Stability of closed-loop systems is of **utmost** importance in control system design.
- For a **stable** system, a **bounded input** should result in a corresponding **bounded output**.
- System stability can be characterized either as **absolute** stability or **relative** stability.
- **Absolute** stability refers to whether the system is **stable** or **unstable**.
- **Relative** stability is a **measure** of the **degree** of stability in stable systems.
- The figure below illustrates the **concept** of stability.



(a) Stable



(b) Unstable



Stability and the System's Poles

- To show the relationship between the poles of the system and the stability, recall the Laplace transform of a transfer function is

$$G(s) = \int_0^{\infty} g(t)e^{-st}dt$$

- Taking the absolute value on both sides gives

$$|G(s)| = \left| \int_0^{\infty} g(t)e^{-st}dt \right| \leq \int_0^{\infty} |g(t)||e^{-st}|dt$$

- Since $|e^{-st}| = |e^{-\sigma t}|$, where $\sigma = \text{Re}(s)$, when s assumes a value of a pole of $G(s)$, $G(s) = \infty$.
- Thus the magnitude equation can be expressed as

$$\infty \leq \int_0^{\infty} |g(t)||e^{-\sigma t}|dt$$

- If any of the poles of the system are in the right-half of the s -plane or on the $j\omega$ axis then

$$|e^{-\sigma t}| \leq M = 1$$

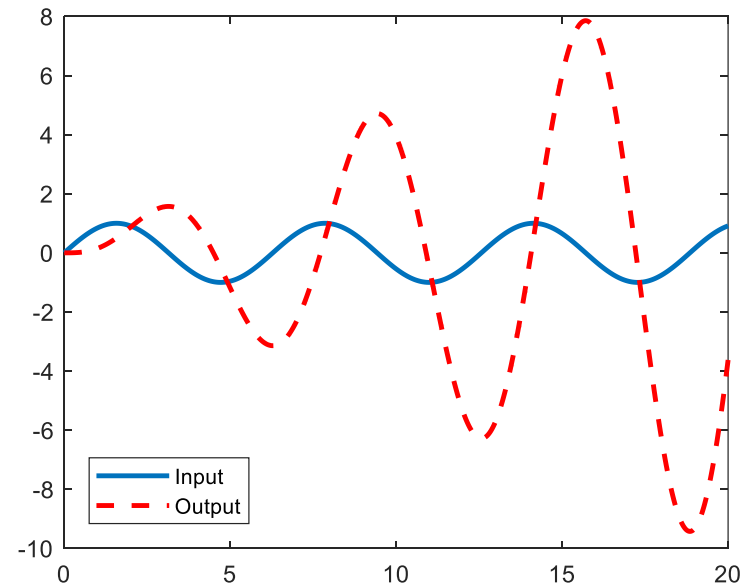


Stability and the System's Poles

- Thus the magnitude equation can be expressed as

$$\infty \leq \int_0^{\infty} M|g(t)|dt = \int_0^{\infty} |g(t)|dt$$

- The above equation clearly violates the **BIBO** stability requirement.
- Thus, a necessary and sufficient condition for a feedback system to be stable is that all the poles of the system have **negative real** parts.
- When a system has **roots** on the **$j\omega$ axis** for example at $s = +j\omega$ and $s = -j\omega$ the system is **marginally** stable.
- This is because, if the input is a sinusoid, of the form $\sin \omega t$ then the output will also be a sinusoid of the form $t \sin \omega t$ which is **unbounded**
- The figure shows the response of a system $G(s) = \frac{1}{s^2+1}$ to a **sine** input.





The Routh Stability Criterion

- The Routh-Hurwitz criterion is a **necessary** and **sufficient** criterion for the stability of **linear** systems.
- The criterion is based on ordering the **coefficients** of the characteristic equation

$$a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \dots + a_1 s + a_0 = 0$$

- The coefficients are arranged into the **Routh array** as below

$$\begin{array}{c|ccc} s^n & a_n & a_{n-2} & a_{n-4} \\ s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} \\ s^{n-2} & b_{n-1} & b_{n-3} & b_{n-5} \\ s^{n-3} & c_{n-1} & c_{n-3} & c_{n-5} \\ \vdots & \vdots & \vdots & \vdots \\ s^0 & h_{n-1} & 0 & 0 \end{array}$$

where

$$b_{n-1} = \frac{a_{n-1} a_{n-2} - a_n a_{n-3}}{a_{n-1}} = \frac{-1}{a_{n-1}} \begin{vmatrix} a_n & a_{n-2} \\ a_{n-1} & a_{n-3} \end{vmatrix}$$

$$b_{n-3} = \frac{-1}{a_{n-1}} \begin{vmatrix} a_n & a_{n-4} \\ a_{n-1} & a_{n-5} \end{vmatrix}, \quad c_{n-1} = \frac{-1}{b_{n-1}} \begin{vmatrix} a_{n-1} & a_{n-3} \\ b_{n-1} & b_{n-3} \end{vmatrix}, \text{ etc}$$



The Routh Stability Criterion

- The algorithm for computing the array entries can proceed on **determinant** basis by using the form of the b_{n-1} equation.
- The Routh-Hurwitz stability criterion states that for the characteristic equation

$$a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \dots + a_1 s + a_0 = 0$$

to have no positive real parts:

- ✓ There is a **necessary**, but **not sufficient** condition that all coefficients have the **same sign** and **none** are **zero**.
- ✓ All coefficients of the first column of the Routh array have the same sign. The **number** of **sign changes** in the **first column** of the Routh array is **equal** to the **number** of positive real **roots**.
- Therefore a **necessary** and **sufficient** condition for a **stable** system is that there be **no changes** in **sign** in the **first column** of the Routh array.



The Routh Stability Criterion: Example

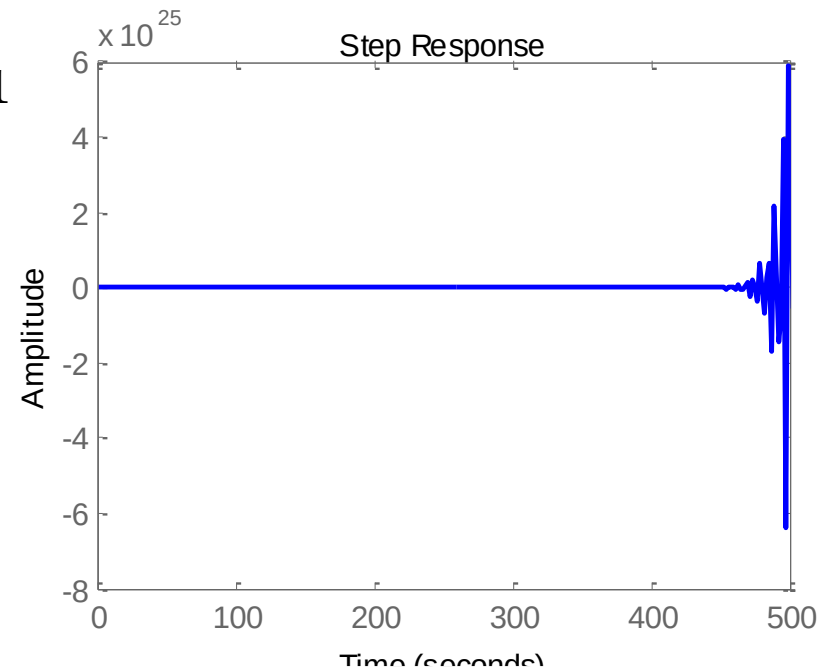
- **Example:** Check the stability of the system with the following characteristic equation

$$s^3 + 2s^2 + 3s + 8 = 0$$

- ✓ All coefficients are present and have the same sign.
- The Routh's array is computed as below

$$\begin{array}{c|cc} s^3 & 1 & 3 \\ s^2 & 2 & 8 \\ s^1 & -1 & 0 \\ s^0 & 8 & 0 \end{array} \quad \begin{array}{l} b_{n-1} = \frac{2(3) - 8(1)}{2} = -1 \\ c_{n-1} = \frac{-1(8)}{-1} = 8 \end{array}$$

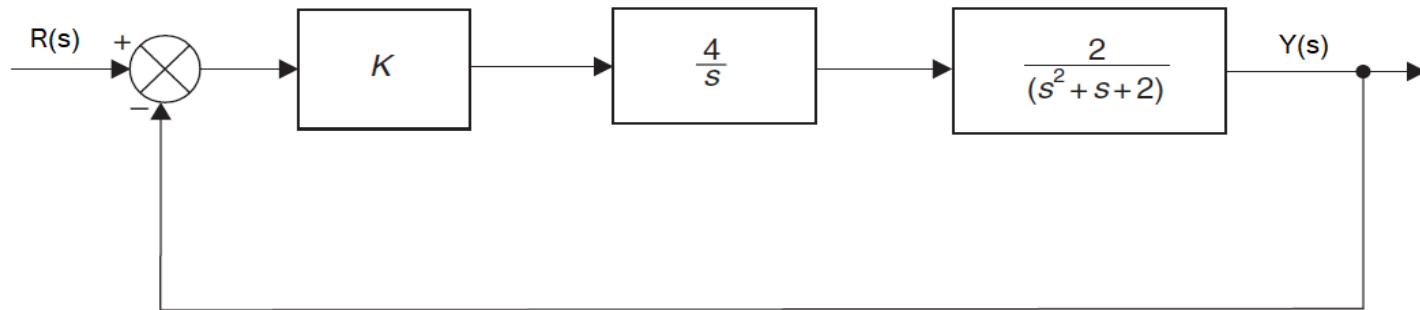
- A **double** sign change in the first column indicates that there are **two** roots with **positive real** parts and hence the system is **unstable** as can be seen in the step response.





Routh's Array: Example 2

- Given the system below, find the value of the gain K to make the system marginally stable.



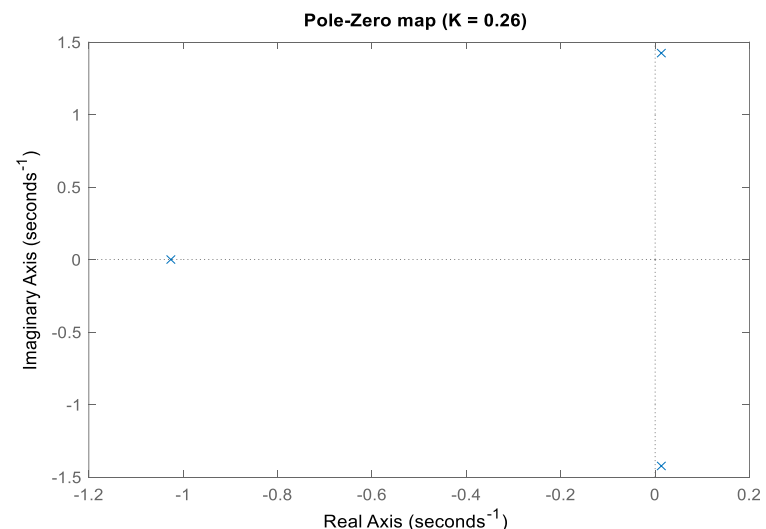
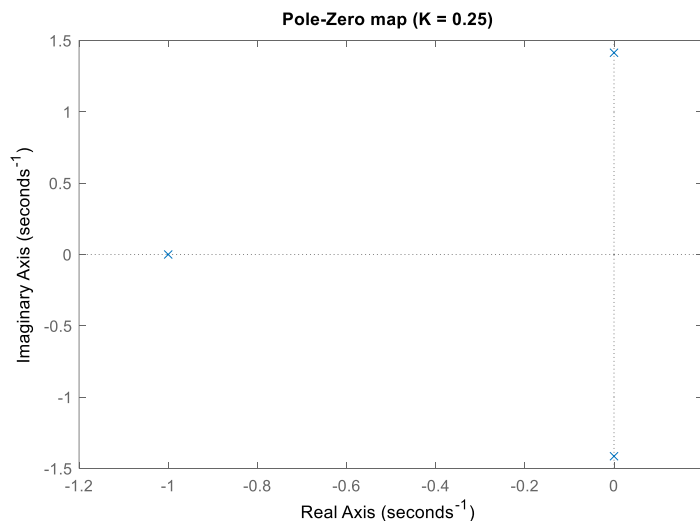
- Take $C(s) = K$, and $G(s) = \frac{8}{s(s^2 + s + 2)}$
- The characteristic equation is given as $1 + C(s)G(s) = 0$ i.e.
$$s^3 + s^2 + 2s + 8K = 0$$
- The Routh's array can thus be populated as below

s^3	$\left \begin{array}{cc} 1 & 2 \end{array} \right.$	$b_{n-1} = \frac{-1}{1}(8K - 2) = 2 - 8K$
s^2	$\left \begin{array}{cc} 1 & 8K \end{array} \right.$	
s^1	$\left \begin{array}{cc} 2 - 8K & 0 \end{array} \right.$	$c_{n-1} = \frac{-1}{2 - 8K}(-8K)(2 - 8K) = 8K$
s^0	$\left \begin{array}{cc} 8K & 0 \end{array} \right.$	



Routh's Array: Example 2

- For a sign change in the first column of the Routh array, either
$$2 - 8K < 0 \text{ or } 8K < 0.$$
- But the gain K cannot be negative thus for marginal stability,
$$2 - 8K = 0 \Rightarrow K = 0.25$$
- When the gain K is 0.25, the characteristic equation becomes
$$s^3 + s^2 + 2s + 2 = 0$$
- The Pole-Zero map (left) shows two roots on the imaginary axis and a slight increase in K results in them crossing to the right of the s -plane (right) thereby rendering the system unstable.



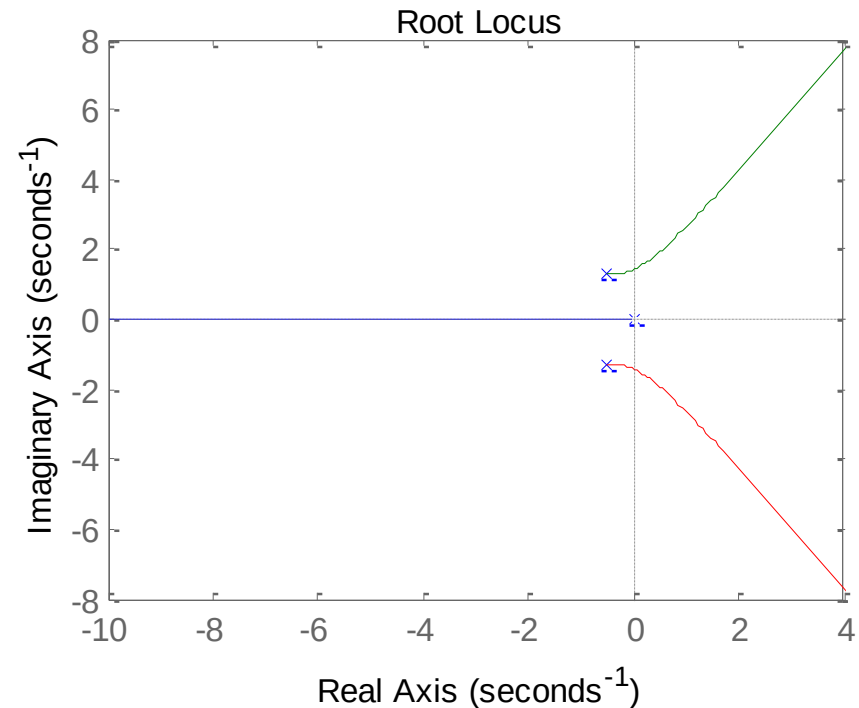


Example 2: Root Locus

- The **root locus** plot shows the trajectories of the system's roots as the gain varies from 0 to ∞ .
- The following MATLAB commands plot the root locus of the system in Example 2.

```
>> G = tf(8*0.001, [1 1 2 0])  
>> rlocus(G)
```

- Note the root locus in MATLAB is plotted using the **open loop** transfer function with a value of the gain k close to zero.
- Clicking on any point on the root locus in the Matlab *Figure* window shows the corresponding gain at that point.





Example 2: Step responses

- The figures show the step responses of the system for $K = 0.25, 0.24$ and 0.26 .
- For $K = 0.25$ the output is sinusoid with constant amplitude (marginally stable).
- When $K < 0.25$ the system is stable and **unstable** for $K > 0.25$.

